

The Riemann Hypothesis and Cryptographic Security: What the Cathedral Proves, What It Doesn't, and What the Headlines Will Get Wrong

Jason Robert Gochanour

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Abstract

We analyze the cryptographic security implications of the Cathedral, a machine-verified Lean 4 formalization establishing that the Riemann Hypothesis is equivalent to the decay of the Nyman–Beurling distance. We rigorously demonstrate that this result—and indeed any resolution of the Riemann Hypothesis, whether proved or disproved—has **no direct impact** on the security of RSA, elliptic curve cryptography, AES, SHA-256, or any deployed cryptographic system. We identify and debunk seven common misconceptions that security researchers and journalists may propagate, provide precise separations between distributional knowledge (which RH provides) and computational hardness (which cryptography requires), and analyze the one scenario in which RH-adjacent mathematics *could* theoretically affect cryptography: the discovery of a new structural property of primes during the proof process. We conclude that the Cathedral is cryptographically neutral, and that proving the Riemann Hypothesis would, if anything, *strengthen* the theoretical foundations of existing cryptographic systems by converting conditional results to unconditional ones.

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1 Introduction: Why This Paper Is Necessary

When media outlets report on progress toward the Riemann Hypothesis, the following headline pattern is nearly inevitable:

“Scientists Crack 167-Year-Old Math Problem — Is Your Encryption at Risk?”

The answer is no. But the reasons are subtle enough that they deserve a rigorous treatment, particularly because the Cathedral involves primes, zeta functions, and L^2 approximation theory — all of which sound superficially connected to cryptography.

This paper serves three audiences:

1. **Security researchers** who need to assess whether the Cathedral warrants a threat advisory.
2. **Journalists** who need accurate language for public reporting.
3. **Cryptographers** who want precise statements about what distributional results (like RH) do and do not imply about computational hardness.

1.1 Summary of Findings

Bottom line: The Cathedral (and any resolution of RH) has zero impact on the security of:

- RSA (factoring-based)
- ECDSA / EdDSA (elliptic curve-based)
- AES-128/256 (symmetric encryption)
- SHA-2 / SHA-3 (hash functions)
- CRYSTALS-Kyber / Dilithium (lattice-based, post-quantum)
- Signal Protocol, TLS 1.3, or any deployed system

2 What the Cathedral Actually Proves

The crown theorem of the Cathedral states:

Theorem 2.1 (Nyman–Beurling Equivalence, Machine-Verified).

$$\text{RH} \iff \forall \varepsilon > 0, \exists N_0, \forall N \geq N_0, \exists v \in \mathbb{R}^{N-1} : \int_0^1 \left(1 - \sum_{k=2}^N v_k \{1/(kx)\} \right)^2 dx < \varepsilon.$$

This is a statement about the density of a function space. It says nothing about:

- Factoring integers.
- Computing discrete logarithms.
- Finding collisions in hash functions.
- Solving lattice problems.

- Any algorithmic task used in cryptography.

The Cathedral is a *structural equivalence*, not an algorithm. It reduces one mathematical statement (RH) to another equivalent mathematical statement ($d_N^2 \rightarrow 0$). Neither statement provides a computational shortcut for any known hard problem.

3 Seven Myths Debunked

Myth 1 (“Proving RH breaks RSA.”). *False.*

RSA security relies on the computational hardness of factoring $n = pq$ where p and q are large primes. Specifically:

Definition 3.1 (Factoring Assumption). There is no classical algorithm that, given a uniformly random n -bit semiprime $N = PQ$, outputs P (or Q) in time polynomial in n , with non-negligible probability.

The Riemann Hypothesis is a statement about the *distribution* of primes: it constrains the error term in the Prime Number Theorem to $O(\sqrt{x} \log^2 x)$ rather than the unconditional $O(x \exp(-c\sqrt{\log x}))$.

Claim 3.2 (Distribution $\neq$ Factoring). Knowing the exact distribution of primes does not help factor a specific semiprime. The analogy: knowing that 50% of the population is female does not help you determine the gender of a specific individual.

More precisely: the prime-counting function $\pi(x)$ tells you *how many* primes exist below x . Factoring requires knowing *which* integers near $\sqrt{N}$ are prime and divide N . These are fundamentally different computational problems. The first is solved by RH (with optimal error bounds); the second is conjectured to be hard regardless of RH.

Remark 3.3. Cryptographers already routinely *assume* RH is true. Many theoretical results in cryptography are proved “conditional on GRH” (the Generalized Riemann Hypothesis). If RH were secretly useful for breaking RSA, someone would have already exploited the assumption.

Myth 2 (“Proving RH reveals a pattern in the primes that could be exploited.”). *False.*

RH constrains the *global statistical distribution* of primes. It tells you that primes of size $\sim x$ occur with density $\sim 1/\ln x$, with fluctuations of size $\sim \sqrt{x}$. This is already assumed in every cryptographic security analysis.

The “pattern” that RH reveals is:

Primes are distributed as regularly as possible, subject to their arithmetic constraints.

This is the *opposite* of a vulnerability. Irregularity in prime distribution would be more concerning for cryptography than regularity. If primes clustered unexpectedly, it might create exploitable structure in cryptographic keys. RH asserts they *don’t* cluster—they’re as well-distributed as possible.

Myth 3 (“The Möbius function in the proof could be used to factor numbers.”). *False.*

The Cathedral uses the Mertens function $M(x) = \sum_{n \leq x} \mu(n)$, where $\mu(n)$ is the Möbius function. Security researchers may note that $\mu(n)$ encodes the factorization of n :

$$\mu(n) = \begin{cases} (-1)^k & \text{if } n \text{ is a product of } k \text{ distinct primes} \\ 0 & \text{if } n \text{ has a squared prime factor} \end{cases}$$

The circularity is immediate: **computing $\mu(n)$ requires knowing the factorization of n .** You cannot use $\mu(n)$ to factor n —you need the factorization to compute $\mu(n)$ in the first place.

The Mertens function $M(x)$ is a *sum* over all $n \leq x$, which encodes statistical information about the distribution of prime factors across all integers, not about any specific integer. It is computable in $O(x^{2/3})$ time without factoring anything (via the Meissel–Mertens method), but this computation provides no information useful for factoring a specific N .

Myth 4 (“The Gram matrix eigenvalues could reveal prime factors.”). **False.**

The Gram matrix G_N has entries $G(j, k) = \int_0^1 \{1/(jx)\} \{1/(kx)\} dx$, and its eigenvalues depend on the arithmetic interaction between integers $2 \leq j, k \leq N$. Security researchers might wonder: could the eigenvalue structure of G_N (for N near a target semiprime) reveal factoring information?

No. The Gram matrix depends on $\gcd(j, k)$ through the Vasyunin formula, but this is the GCD of the *row/column indices*, not of the target number. The matrix G_N is a fixed mathematical object that depends only on N , not on any secret key or private value. It is publicly computable. There is nothing to “reveal.”

Myth 5 (“The zeta zeros are like private keys—knowing them breaks the system.”). **False.**

The nontrivial zeros of $\zeta(s)$ are public mathematical constants. The first 10^{13} zeros have been computed and published. Databases of zeta zeros are freely available online (LMFDB, Odlyzko’s tables). Knowing billions of zeta zeros has not helped anyone factor any specific integer.

The zeros encode the *Fourier spectrum* of the prime counts, not the location of individual primes. They are the frequency content of the signal $\pi(x) - \text{Li}(x)$, which tells you about the global oscillation of the prime-counting function, not about specific values.

Myth 6 (“Disproving RH would be worse for cryptography.”). **Also false** (*with a nuance*).

If RH were *disproved*—if a zero were found off the critical line at $\rho = \sigma + it$ with $\sigma \neq 1/2$ —the consequences would be:

1. The prime-counting function $\pi(x)$ would have larger fluctuations than currently assumed: $O(x^\sigma)$ instead of $O(x^{1/2})$.
2. Certain conditional results in complexity theory (proved assuming GRH) would lose their theoretical backing.
3. The Mertens bound $|M(x)| = O(x^{1/2+\varepsilon})$ would fail, replaced by $|M(x)| = O(x^{\sigma+\varepsilon})$ for $\sigma > 1/2$.

None of these affect factoring hardness. The security of RSA does not depend on the size of $\pi(x) - \text{Li}(x)$; it depends on the hardness of factoring specific integers.

The nuance: a disproof would mean that primes can cluster or gap more than expected. In principle, this could affect the output distribution of prime-generation algorithms (e.g., the probability that a random odd number in $[2^{1023}, 2^{1024}]$ is prime). But in practice, the error term is already negligible: for 1024-bit primes, the density is $\sim 1/710$ regardless of RH, and the fluctuation term is of order $2^{512}/710^2 \approx 10^{-6}$ —far too small to exploit.

Myth 7 (“The proof technique itself (Parseval Bridge, Mellin transform) could be weaponized.”). **False.**

The Parseval Bridge converts an L^2 norm on $(0, 1)$ to an L^2 norm on the critical line. The Mellin transform connects time-domain and frequency-domain representations. These are standard tools in harmonic analysis and signal processing. They are already used in cryptanalysis (e.g., the Number Field Sieve uses lattice reduction, which involves Fourier analysis). The Cathedral does not introduce any new algorithmic technique—it provides a new *structural understanding* of a known equivalence.

Structural understanding is valuable for mathematics but not directly useful for algorithm design. Knowing *why* $d_N^2 \rightarrow 0$ is equivalent to RH does not tell you *how* to factor integers.

4 The Precise Separation: Distribution vs. Computation

The fundamental reason that RH does not affect cryptography is a gap between two types of mathematical knowledge:

Definition 4.1 (Distributional Knowledge). A statement about the *statistical properties* of a class of numbers. Example: “There are $\sim x / \ln x$ primes below x .”

Definition 4.2 (Computational Knowledge). A statement about the *algorithmic difficulty* of a specific task on a given input. Example: “Factoring $N = PQ$ requires $\exp(\Omega(n^{1/3}))$ time for n -bit N .”

The Riemann Hypothesis is purely distributional:

What RH tells you	What RH does NOT tell you
How many primes $\leq x$	Which integers are prime
Error in prime counting	How to factor $N = PQ$
$ M(x) \leq C\sqrt{x} \log^2 x$	$\mu(N)$ for specific N
$\zeta(s) \neq 0$ for $\text{Re } s > 1/2$	Discrete log of $g^x \bmod p$
Primes are well-distributed	Private key from public key

Cryptographic hardness assumptions (factoring, discrete log, LWE) are *computational* statements. They assert that certain functions are hard to invert, not that certain statistical distributions have particular shapes. RH constrains the shape; cryptography needs the hardness. These are orthogonal.

4.1 A Formal Argument

Theorem 4.3 (RH Does Not Imply Factoring). There is no known reduction from the Integer Factoring Problem to any decision problem that is resolved by the Riemann Hypothesis. That is: no algorithm is known that, given oracle access to an RH oracle (a function that returns “true” for RH), can factor integers in polynomial time.

Argument. An “RH oracle” is a single-bit oracle: it returns 1 (if RH is true) or 0 (if RH is false). A single bit of information cannot provide a polynomial-time speedup for factoring, since the factoring problem requires $\Omega(n)$ bits of output (the factor itself has $n/2$ bits).

More generally: RH is a single mathematical statement. Its truth value is a fixed constant (either $\top$ or $\perp$). Any algorithm can be modified to hardcode this constant with at most a constant factor overhead. If “assuming RH” gave a factoring algorithm, then the unconditional algorithm (which simply tries both branches) would also be polynomial-time. $\square$

Remark 4.4. This argument generalizes: *no fixed mathematical theorem can break a cryptosystem*, because the theorem’s content can be hardcoded. Only *algorithmic techniques* can break cryptosystems, and a theorem statement (however deep) is not an algorithm.

5 What *Could* (Theoretically) Matter

While the Cathedral itself is cryptographically neutral, intellectual honesty requires addressing the one scenario that *could* theoretically affect cryptography:

5.1 The Technique Spillover Hypothesis

The process of *proving* a deep theorem sometimes produces new mathematical techniques that have unexpected applications. For example:

- Wiles’ proof of Fermat’s Last Theorem developed the theory of Galois representations, which later influenced elliptic curve cryptography (both positively and as an analytical tool).
- Shor’s algorithm for quantum factoring uses the Quantum Fourier Transform, which is a general-purpose tool with many applications.

If a future proof of RH required developing a fundamentally new technique for understanding prime factorizations (not just distributions), that technique *might* have algorithmic applications. This is speculative and has no evidence base.

Claim 5.1 (Cathedral Technique Assessment). The techniques used in the Cathedral—both active and archived—are:

1. Nyman–Beurling L^2 density (1950s).
2. Báez-Duarte basis (2003).
3. Mellin transform and Plancherel theorem (19th century).
4. Abel summation (17th century).
5. Montgomery–Vaughan mean value theorem (1973).
6. Rayleigh–Ritz variational principle (19th century).
7. Sieve methods: Selberg sieve, Vaughan decomposition (1940s–70s).
8. Perron’s formula for Dirichlet series (1908).
9. Vasyunin cotangent formula and Dedekind sums (1892–1995).
10. Robin’s inequality and the divisor function (1984).

All of these are well-known, decades-old techniques in harmonic analysis and analytic number theory. None of them provide any avenue for algorithmic improvement in factoring, discrete log, or lattice problems. The Cathedral does not introduce new mathematical techniques—it assembles existing ones in a new architecture that is machine-verified.

5.2 The GRH-Conditional Theorems

Several results in theoretical computer science are proved assuming the Generalized Riemann Hypothesis (GRH), which is stronger than RH:

GRH-Conditional Result	Impact if GRH Proved
Miller’s deterministic primality test	Unconditional $O(\log^4 n)$
Polynomial-time class group computation	Unconditional
Smallest quadratic non-residue $\leq O(\log^2 p)$	Unconditional
Bach’s bounds for generating primes	Tighter constants

If GRH were proved, these conditional results would become unconditional. None of them break any cryptosystem:

- **Miller’s test:** Primality testing is already polynomial-time unconditionally (AKS, 2002). The GRH-conditional Miller test is faster but solves the same problem. Making primality testing slightly faster does not help factoring.
- **Class group computation:** Used in computational algebraic number theory, not in attacks on deployed systems.
- **Quadratic non-residues:** The smallest quadratic non-residue modulo p is already known to be $O(\sqrt{p} \log p)$ unconditionally (Pólya–Vinogradov). The GRH improvement to $O(\log^2 p)$ does not provide an attack vector.

Remark 5.2 (GRH Actually Helps Security Proofs). Several cryptographic security proofs *assume* GRH. If GRH were proved, these proofs would become unconditional, making the security guarantees *stronger*, not weaker. Examples include certain proofs about the Goldwasser–Micali encryption scheme and the Blum–Blum–Shub pseudorandom generator.

6 Post-Quantum Considerations

Modern cryptography is transitioning to post-quantum algorithms (NIST PQC standards: CRYSTALS-Kyber, CRYSTALS-Dilithium, SPHINCS+). These are based on:

- Lattice problems (Learning with Errors, LWE).
- Hash-based signatures.
- Code-based cryptography.

None of these have any mathematical connection to the Riemann zeta function, prime distribution, or the Nyman–Beurling criterion. The hardness of LWE is related to worst-case lattice problems (Shortest Vector Problem), which live in a completely different branch of mathematics (geometry of numbers, rather than analytic number theory).

Claim 6.1 (Post-Quantum Immunity). The Cathedral’s result is irrelevant to post-quantum cryptographic security. The NIST PQC standards are not affected by any resolution of the Riemann Hypothesis.

The actual threat to pre-quantum cryptography (RSA, ECC) is **Shor’s algorithm**, which factors integers and computes discrete logarithms in quantum polynomial time. Shor’s algorithm has nothing to do with the Riemann Hypothesis—it uses the quantum Fourier transform to find the period of modular exponentiation, which is a group-theoretic technique, not a number-theoretic one.

7 Archive Audit: 96 Abandoned Approaches

The Cathedral preserves 96 archived Lean files documenting approaches that were explored and abandoned during the 23-day construction. A natural question arises: could any of these abandoned techniques be weaponized, even if the main proof architecture cannot?

We audited all 96 archived files, categorized by technique:

Category	Technique	Files	Finding
Sieve methods	Selberg, Vaughan, parity barrier	5	No threat
Perron/contour	Contour shifts, Perron kernel	4	No threat
Spectral theory	Eigenvalues, Hilbert–Pólya	3	No threat
Robin’s inequality	$\sigma(n) < e^\gamma n \ln \ln n$	6	No threat
Vasyunin/cotangent	Dedekind sums, piecewise FTC	13	No threat
Mellin transforms	Floor Mellin, Dirichlet series	6	No threat
Exotic approaches	Bessel, winding number, QFT	18	No threat
Duplicate/structural	Superseded declarations	41	No threat

7.1 The Archive as Counter-Evidence

The archive *strengthens* the security assessment. Each archived file documents an approach that was explored and *failed*. These 96 failures constitute concrete evidence that the mathematical tools used in the Cathedral do not produce computational shortcuts.

Claim 7.1 (Archive as Counter-Evidence). If sieve methods, Perron’s formula, or Mellin transform techniques could be repurposed for integer factoring or discrete logarithm computation, the archived approaches that used these techniques would have *succeeded*—they would have produced unexpected algorithmic results instead of mathematical dead ends. Their failure is evidence *for* the distributional-vs-computational separation (Section 4).

The distinction is important: these are not hypothetical arguments about what these tools *might* do. They are empirical records of what these tools *actually did* when applied to a concrete problem. The answer: they bounded L^2 norms, computed Gram matrix entries, and controlled Dirichlet polynomial averages. At no point did any approach produce information about the factorization of specific integers.

7.2 The Rust Experiments (211,202 lines)

The companion Rust codebase computes Gram matrix entries, eigenvalues, and Rayleigh quotients using exact rational arithmetic for $N \leq 5000$. It processes the integers $2, 3, \dots, N$ using gcd and lcm operations. This is publicly reproducible computation on small public integers. No private keys, secrets, or cryptographic targets are involved at any point in the numerical pipeline.

8 Symmetric Cryptography and Hash Functions

For completeness: symmetric cryptography (AES, ChaCha20) and hash functions (SHA-2, SHA-3, BLAKE3) have no connection to prime numbers or number theory whatsoever. Their security is based on:

- Confusion and diffusion (Shannon, 1949).
- Resistance to linear and differential cryptanalysis.
- The random oracle model (for hash functions).

The Riemann Hypothesis could be proved, disproved, or shown to be independent of ZFC, and AES-256 would remain equally secure (or insecure) as it is today.

9 What a Responsible Disclosure Looks Like

Since the Cathedral is cryptographically neutral, no security advisory is warranted. However, for the benefit of responsible disclosure processes, we state explicitly:

1. **No CVE is needed.** The Cathedral does not disclose a vulnerability in any system.
2. **No key material is compromised.** The proof involves no private keys, secrets, or confidential data.
3. **No algorithm is accelerated.** The proof does not provide a faster factoring, discrete log, or lattice algorithm.
4. **No system requires patching.** No software update is needed in response to this work.
5. **No embargo is appropriate.** The work is pure mathematics and should be published openly.

10 A Note for Journalists

If you are writing about the Cathedral for a general audience, here are accurate and inaccurate framings:

Accurate	Inaccurate
“Researchers built a computer-verified proof about the oldest unsolved problem in math.”	“Researchers cracked the code behind internet encryption.”
“The result translates the Riemann Hypothesis into an equivalent form, but does not solve it.”	“The Riemann Hypothesis has been solved / nearly solved.”
“This work has no impact on cryptographic security.”	“Your passwords / bank accounts may be at risk.”
“The proof is about the statistical distribution of prime numbers.”	“The proof reveals a pattern in primes that hackers could exploit.”
“If anything, proving RH would strengthen crypto security proofs.”	“Proving RH would break the internet.”

11 Conclusion

The Cathedral is a machine-verified proof that the Riemann Hypothesis is equivalent to the decay of the Nyman–Beurling distance. It is a contribution to pure mathematics and proof engineering, with no direct, indirect, or foreseeable implications for cryptographic security.

The disconnect between RH and cryptography is not accidental—it reflects a deep structural separation between *distributional* properties of primes (which RH constrains) and *computational* hardness assumptions (which cryptography requires). These are different mathematical universes. Progress in one does not translate to the other.

If the Riemann Hypothesis is eventually proved:

- Mathematicians will celebrate.
- Cryptographers will note that several conditional security proofs are now unconditional (a mild strengthening).

- Security practitioners will change nothing.
- Your bank account will be fine.

If the Riemann Hypothesis is eventually disproved:

- Mathematicians will be astonished.
- Cryptographers will note that some error bounds in prime generation are slightly looser than assumed (negligible in practice).
- Security practitioners will *still* change nothing.
- Your bank account will *still* be fine.

The primes are not the lock. The primes are the landscape. Knowing the landscape perfectly does not give you the key.

Sleep well. Your encryption is safe.

Source code: <https://github.com/jrgochan/prime>

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